



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For



Project Management Professional (PMP) Exam Prep

Course Code: C523

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version v2.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	20 July 2026	Initial release of the PMP Exam Prep courseware.	Dr. Alfred Ang
2.0	20 July 2026	Full-depth build: the deck now carries the complete ECO 2026 teaching content across all six topics (536 slides), with per-topic concept, process, formula, ITTO and exam-focus slides. Day 4 afternoon is given over to the capstone lab and the self-scored mock practice exam.	Dr. Alfred Ang

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Course Overview

Project Management Professional (PMP) Exam Prep is a 4-day, 32-hour instructor-led programme delivering the 35 contact hours required for the PMI Project Management Professional (PMP)® application.

The course is fully aligned to the PMI PMP Examination Content Outline published for the July 2026 exam update. That update rebalanced the exam domains and introduced sustainability and artificial intelligence as examinable considerations:

- Domain People — 33% of exam items. How you lead, align and communicate with the humans on and around the project.
- Domain Process — 41% of exam items. How you plan, deliver, measure and close the work itself.
- Domain Business Environment — 26% of exam items. How the project connects to governance, compliance, risk and organisational change.

Approach coverage: About 40% predictive; the remaining 60% split between adaptive/agile and hybrid.

Target Audience

- Project managers and project leads responsible for medium-scale projects
- Team leads, engineers and analysts moving into project management
- PMO staff, business analysts and consultants supporting project delivery
- Candidates preparing to sit the PMP certification examination

Prerequisites

- Working familiarity with a project environment, in any industry
- No prior PMP study is assumed; all concepts are taught from first principles
- A laptop with a spreadsheet application and a modern web browser

Course Details

Course Reference	C523
Trainer	Dr. Alfred Ang
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	4 days / 32 hours
PDU Value	35 PDUs / contact hours

Learning Outcomes

By the end of this course, learners will be able to:

Ref	Learning Outcome	Reinforced by
LO1	Scope medium-scale project requirements to drive timely completions.	Labs 02–06, 09–11 — environment scan, business case, charter, requirements traceability and WBS
LO2	Develop project plans based on realistic timelines and resource allocations.	Labs 11–13, 15, 20 — WBS, critical path with PERT, cost baseline with reserves, resource plan and schedule compression
LO3	Implement methodologies effectively to address project exigencies and derive solutions from identified root causes.	Labs 08, 16, 17, 21 — team charter, Tuckman diagnosis, conflict role-play and root cause analysis
LO4	Analyze program risks and engage stakeholders through scheduled touchpoints to discuss potential issues.	Labs 04, 07, 14, 18 — compliance register, stakeholder analysis, risk register with EMV and the communication plan
LO5	Coordinate project deliverables against set objectives, costs, and timelines, and implement corrective actions.	Labs 19, 20, 22, 23 — flow metrics, earned value, control charts, closure and benefits realisation

Examination Content Outline Coverage

Every task in the July 2026 Examination Content Outline is delivered within one of the six course topics. The mapping below is the coverage evidence for that claim.

Topic	ECO Domain & Tasks Delivered	Day
Topic 1. Business Environment	Business Environment T1: Define and establish project governance; Business Environment T2: Plan and manage project compliance; Business Environment T7: Support organizational change; Business Environment T8: Evaluate external business environment changes; Process T3: Help ensure value-based delivery	Day 1
Topic 2. Start the Project	People T1: Develop a common vision; People T4: Engage stakeholders; People T5: Align stakeholder expectations; People T6: Manage stakeholder expectations; Process T1: Develop an integrated project management plan and plan delivery	Day 1
Topic 3. Plan the Project	Process T1: Develop an integrated project management plan and plan delivery; Process T2: Develop and manage project scope; Process T4: Plan and manage resources; Process T5: Plan and manage procurement; Process T6: Plan and manage finance; Process T7: Plan and optimize quality of products/deliverables; Process T8: Plan and manage schedule; Business Environment T5: Plan and manage risk; People T8: Plan and manage communication	Day 2
Topic 4. Lead the Project Team	People T1: Develop a common vision; People T2: Manage conflicts; People T3: Lead the project team; People T7: Help ensure knowledge transfer; People T8: Plan and manage communication; Process T1: Develop an integrated project management plan and plan delivery	Day 3
Topic 5. Monitor and Control the Project	Process T9: Evaluate project status; Business Environment T3: Manage and control changes; Business Environment T4: Remove impediments and manage issues; Business Environment T5: Plan and manage risk; Business Environment T6: Continuous improvement; Process T7: Plan and optimize quality of products/deliverables	Day 3
Topic 6. Close the Project	Process T10: Manage project closure; Business Environment T6: Continuous improvement; People T7: Help ensure knowledge	Day 4

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Day 1: Business Environment & Starting the Project

Topics covered: Topic 1. Business Environment, Topic 2. Start the Project

Hands-on labs: Lab 01 PMP Exam Orientation and 35-PDU Study Plan, Lab 02 PESTLE and TECOP External Environment Scan, Lab 03 EEF/OPA Inventory with Governance and Escalation Thresholds, Lab 04 Compliance, Sustainability and AI Governance Register, Lab 05 Business Case and Cost-Benefit Analysis, Lab 06 Project Charter, Lab 07 Stakeholder Register, Power/Interest Grid and Salience Model, Lab 08 Team Charter, Ground Rules and Shared Vision

Time	Topic / Session	Activity	Duration
9:30 – 10:00	Registration & Course Opening	Course introduction, trainer and learner introductions, ground rules, learning outcomes, ECO 2026 orientation	30 min
10:00 – 11:15	Topic 1. Business Environment	Concept delivery, worked examples and discussion	75 min
11:15 – 11:30	Tea Break	—	15 min
11:30 – 13:00	Topic 2. Start the Project	Concept delivery, worked examples and discussion	90 min
13:00 – 14:00	Lunch Break	—	60 min
14:00 – 15:45	Hands-on Labs	Lab 01, Lab 02, Lab 03, Lab 04 (LO1, LO3, LO4)	105 min
15:45 – 16:00	Tea Break	—	15 min
16:00 – 17:30	Hands-on Labs (continued)	Lab 05, Lab 06, Lab 07, Lab 08 — group share-back	90 min
17:30 – 18:30	Consolidation	Exam-style question drill on the day's material, recap and Q&A	60 min

Total instructional time: 480 minutes (8 hours), excluding the 1-hour lunch break.

Day 2: Planning the Project

Topics covered: Topic 3. Plan the Project

Hands-on labs: Lab 09 Requirements Elicitation and Traceability Matrix, Lab 10 MoSCoW and Kano Prioritisation with Product Backlog, Lab 11 Work Breakdown Structure and WBS Dictionary, Lab 12 Network Diagram, PERT Estimating and Critical Path, Lab 13 Cost Estimating, Budget and Reserve Analysis, Lab 14 Risk Register, Probability/Impact Matrix and EMV Decision Tree

Time	Topic / Session	Activity	Duration
9:30 – 10:00	Registration & Course Opening	Recap of Day 1, exam-style warm-up questions and Q&A	30 min
10:00 – 11:15	Topic 3. Plan the Project	Concept delivery, worked examples and discussion	75 min
11:15 – 11:30	Tea Break	—	15 min
11:30 – 13:00	Topic 3. Plan the Project	Concept delivery, worked examples and discussion	90 min
13:00 – 14:00	Lunch Break	—	60 min
14:00 – 15:45	Hands-on Labs	Lab 09, Lab 10, Lab 11 (LO1, LO2, LO4)	105 min
15:45 – 16:00	Tea Break	—	15 min
16:00 – 17:30	Hands-on Labs (continued)	Lab 12, Lab 13, Lab 14 — group share-back	90 min
17:30 – 18:30	Consolidation	Exam-style question drill on the day's material, recap and Q&A	60 min

Total instructional time: 480 minutes (8 hours), excluding the 1-hour lunch break.

Day 3: Leading the Team & Controlling Delivery

Topics covered: Topic 4. Lead the Project Team, Topic 5. Monitor and Control the Project

Hands-on labs: Lab 15 RACI Responsibility Assignment Matrix, Lab 16 Tuckman Diagnosis and Motivation Theory Application, Lab 17 Conflict Resolution Role-Play across the Five Modes, Lab 18 Communication Plan, Channels Formula and Status Report, Lab 19 Kanban Board with WIP Limits and Lead/Cycle Time, Lab 20 Earned Value Management and Schedule Compression, Lab 21 Root Cause Analysis with 5 Whys, Fishbone and Pareto, Lab 22 SPC Control Chart and Statistical Process Analysis

Time	Topic / Session	Activity	Duration
9:30 – 10:00	Registration & Course Opening	Recap of Day 2, exam-style warm-up questions and Q&A	30 min
10:00 – 11:15	Topic 4. Lead the Project Team	Concept delivery, worked examples and discussion	75 min
11:15 – 11:30	Tea Break	—	15 min
11:30 – 13:00	Topic 5. Monitor and Control the Project	Concept delivery, worked examples and discussion	90 min
13:00 – 14:00	Lunch Break	—	60 min
14:00 – 15:45	Hands-on Labs	Lab 15, Lab 16, Lab 17, Lab 18 (LO2, LO3, LO4, LO5)	105 min
15:45 – 16:00	Tea Break	—	15 min
16:00 – 17:30	Hands-on Labs (continued)	Lab 19, Lab 20, Lab 21, Lab 22 — group share-back	90 min
17:30 – 18:30	Consolidation	Exam-style question drill on the day's material, recap and Q&A	60 min

Total instructional time: 480 minutes (8 hours), excluding the 1-hour lunch break.

Day 4: Closing, Exam Strategy & Practice

Topics covered: Topic 6. Close the Project

Hands-on labs: Lab 23 Project Closure, Benefits Realisation and Lessons Learned, Lab 24

Capstone: Consolidated Project Plan and Mock Exam

Time	Topic / Session	Activity	Duration
9:30 – 9:45	Day 4 Opening	Recap of Day 3, exam-style warm-up questions and Q&A	15 min
9:45 – 11:00	Topic 6. Close the Project	Concept delivery: closure, benefits realisation, knowledge transfer	75 min
11:00 – 11:15	Tea Break	—	15 min
11:15 – 13:00	Hands-on Labs	Lab 23 (LO1, LO2, LO3, LO4, LO5)	105 min
13:00 – 14:00	Lunch Break	—	60 min
14:00 – 15:45	Capstone Lab	Lab 24 Part A — consolidate all 23 labs' artifacts into one integrated project management plan with a consistency audit	105 min
15:45 – 16:00	Tea Break	—	15 min
16:00 – 17:10	Mock Practice Exam	Lab 24 Part B — 30-question ECO 2026-format mock exam under timed conditions (self-scored, not an assessment)	70 min
17:10 – 18:00	Mock Exam Debrief	Score by domain against the 33/41/26 weights, work the rationale for every wrong answer, and build a dated remediation plan	50 min
18:00 – 18:30	Exam Strategy & Course Close	PMI application steps, study plan, exam-day tactics and Q&A	30 min

Total instructional time: 480 minutes (8 hours), excluding the 1-hour lunch break.

Hands-on Laboratory Schedule

The course includes 24 hands-on labs built around a single running case study, the Contoso Training Portal Upgrade. Each lab consumes artifacts produced by earlier labs, so the set forms one continuous project.

Each lab is timetabled for its core steps in class (820 minutes across the four days — Day 1 180, Day 2 180, Day 3 180, Day 4 280). The extension steps — a further 565 minutes — are completed as self-study after the session, so learners leave with a complete set of project artifacts.

Lab	Title	Topic	ECO Task	LO	In class	Self-study
01	PMP Exam Orientation and 35-PDU Study Plan	T1	All domains	LO1	20 min	25 min
02	PESTLE and TECOP External Environment Scan	T1	BE T8	LO1	20 min	25 min
03	EEF/OPA Inventory with Governance and Escalation Thresholds	T1	BE T1	LO1	20 min	25 min
04	Compliance, Sustainability and AI Governance Register	T1	BE T2	LO4	20 min	25 min
05	Business Case and Cost-Benefit Analysis	T2	Process T3	LO1	25 min	25 min
06	Project Charter	T2	Process T1	LO1	20 min	25 min
07	Stakeholder Register, Power/Interest Grid and Salience Model	T2	People T4, T5	LO4	25 min	25 min
08	Team Charter, Ground Rules and Shared Vision	T2	People T1, T2	LO3	30 min	15 min
09	Requirements Elicitation and Traceability Matrix	T3	Process T2	LO1	30 min	20 min
10	MoSCoW and Kano Prioritisation with Product Backlog	T3	Process T3	LO1	30 min	20 min
11	Work Breakdown	T3	Process T2	LO1	30 min	20 min

	Structure and WBS Dictionary					
12	Network Diagram, PERT Estimating and Critical Path	T3	Process T8	LO2	35 min	25 min
13	Cost Estimating, Budget and Reserve Analysis	T3	Process T6	LO2	25 min	25 min
14	Risk Register, Probability/Impact Matrix and EMV Decision Tree	T3	BE T5	LO4	30 min	30 min
15	RACI Responsibility Assignment Matrix	T4	People T3, Process T4	LO2	20 min	25 min
16	Tuckman Diagnosis and Motivation Theory Application	T4	People T3	LO3	20 min	25 min
17	Conflict Resolution Role-Play across the Five Modes	T4	People T2	LO3	20 min	30 min
18	Communication Plan, Channels Formula and Status Report	T4	People T8	LO4	20 min	30 min
19	Kanban Board with WIP Limits and Lead/Cycle Time	T5	Process T9	LO5	20 min	25 min
20	Earned Value Management and Schedule Compression	T5	Process T9	LO5	25 min	35 min
21	Root Cause Analysis with 5 Whys, Fishbone and Pareto	T5	BE T4	LO3	25 min	35 min
22	SPC Control Chart and Statistical Process Analysis	T5	Process T7	LO5	30 min	30 min

23	Project Closure, Benefits Realisation and Lessons Learned	T6	Process T10, BE T6	LO5	105 min	0 min
24	Capstone: Consolidated Project Plan and Mock Exam	T6	All domains	LO1-LO5	175 min	0 min

Tools and Resources

Learners use the following browser-based tools during the labs. No installation is required and all tools remain available after the course.

Tool	Used for	URL
5 Whys	Drill from a symptom to its root cause by asking 'why' five times.	https://alfredang.github.io/5whys/
Fishbone Diagram	Group candidate causes into categories around a problem statement.	https://alfredang.github.io/fishbone/
Pareto Chart	Rank defect categories to find the vital few driving most of the pain.	https://alfredang.github.io/paretochart/
System Thinking	Map reinforcing and balancing loops behind a recurring problem.	https://alfredang.github.io/systemloop/
SPC / Control Chart	Plot a process over time against control limits to separate signal from noise.	https://alfredang.github.io/novaspcc/
Statistics	Descriptive statistics, distributions and hypothesis testing on project data.	https://alfredang.github.io/novastats/
RACI Matrix	Assign Responsible, Accountable, Consulted and Informed per work package.	https://alfredang.github.io/raci/
Kanban Board	Visualise work in progress, limit WIP and expose bottlenecks.	https://alfredang.github.io/kanban/
Scrum Simulator	Run sprint planning, review and retrospective mechanics.	https://alfredang.github.io/scrum/
Mind Mapping	Expand a central idea into branches for requirements discovery.	https://alfredang.github.io/mindmapping/
Design Thinking	Empathise, define, ideate, prototype and test a solution.	https://alfredang.github.io/designthinking/
Pivot Analysis	Slice project data by dimension to expose trends and outliers.	https://alfredang.github.io/novapivot/

Learning Reinforcement

This is a commercial short course and carries no assessment. Learning is reinforced continuously through the hands-on labs, the daily exam-style drills, and a self-scored mock practice exam on Day 4 that tells each learner where to focus their remaining study.

Mechanism	Format	Covers	When
Hands-on labs	24 labs on one continuous case study, each producing an artifact the next lab consumes	All six topics and every ECO 2026 domain	Throughout, ~21 hours
Lab verification	Each lab states a success test the learner checks their own work against	The technique taught in that lab	End of each lab
Exam-style drills	Scenario questions in the ECO 2026 formats, worked as a group	The day's material	End of Days 1–3
Mock practice exam	30 questions in ECO 2026 format under a timed condition, self-scored by domain	All three domains against the 33/41/26 weights	Day 4, Lab 24 Part B
Domain score analysis	Per-domain percentage compared to the published weights, with a dated remediation plan	The learner's weakest domains	Day 4 debrief

Trainer Notes

- Circulate during every lab — the labs are where misconceptions surface, not during concept delivery.
- Insist learners work the arithmetic by hand; the exam provides no spreadsheet.
- Before answering any scenario, have learners state whether the context is predictive, adaptive or hybrid.
- The Day 4 mock is diagnostic, not a test — frame it that way so learners answer honestly rather than defensively.
- Debrief every wrong answer by rationale; that is where the learning converts.
- Remind learners this course supplies the 35 contact hours PMI requires for the PMP application.